KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION

CHAPTER 4

YEAR 10 MATHEMATICS

Statistics

7 lessons · two pages each · Easy, then Medium

4.1 Types of data
4.2 Frequency tables
4.3 Histograms
4.4 Mean, median, mode and range
4.5 Stem-and-leaf plots
4.6 Symmetrical or skewed
4.7 Time-series graphs

NAME _____ CLASS _____ TEACHER _____

Artwork: a band of gold triple spirals on the edge of the lower bulb, coloured in gold and green, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **two pages**. Read the **learning intention** and tick each **success criterion** when you can do it.

2. Page 1 is **Easy** and page 2 is **Medium**. Each page starts with an example, then questions for you.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. Work down the box one round at a time. Answers go on the line; show working in the boxes.

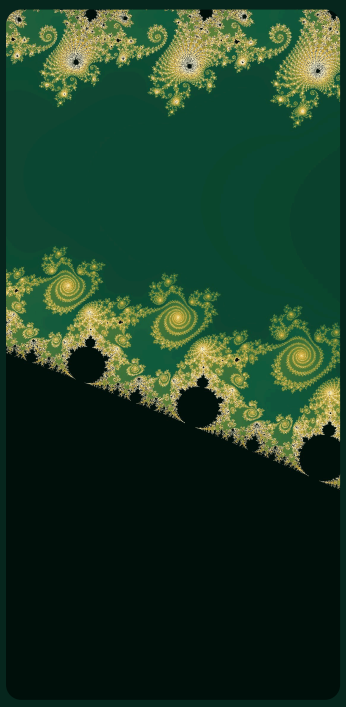
3. Stuck? Read the **Remember** line at the top of the blue box, then look back at the example you copied.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
4.1	Types of data	3		☐
4.2	Frequency tables	5		☐
4.3	Histograms	3		☐
4.4	Mean, median, mode and range	9		☐
4.5	Stem-and-leaf plots	11		☐
4.6	Symmetrical or skewed	13		☐
4.7	Time-series graphs	15		☐

In this chapter you will

- sort data into categorical and numerical, and name each type
- organise data in frequency tables with class intervals and percentage frequency
- read and draw histograms
- find the mean, median, mode and range
- read and draw stem-and-leaf plots
- describe data as symmetrical or skewed
- read time-series graphs and describe trends



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a band of gold triple spirals on the edge of the lower bulb, coloured in gold and green. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 4.3

Histograms

Date _____

Learning intention
We are learning to read a histogram, and draw one from a frequency table.

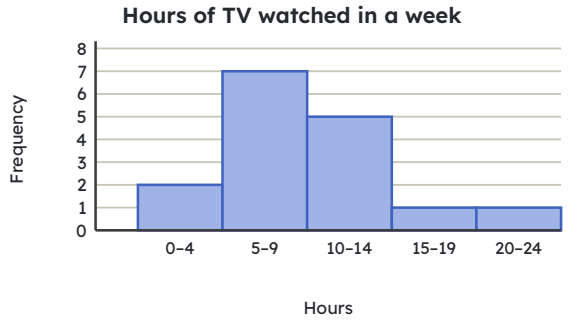
Success criteria
- ☐ I can read the frequency of a class from a histogram
- ☐ I can find the most common class and the total
- ☐ I can draw a histogram and find percentage frequencies

Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 1 A histogram is a column graph for class intervals. Its columns touch.



a How many people watched 5–9 hours?

b Which class is the most common?

c How many people were surveyed?

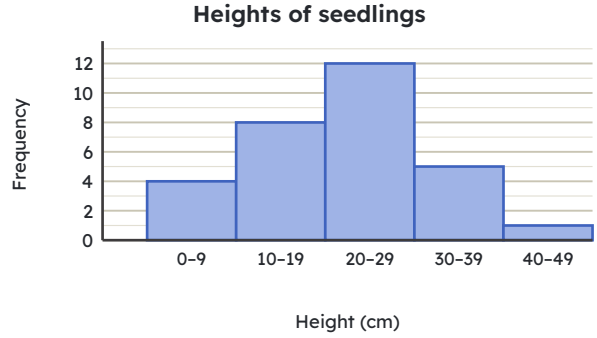
d How many people watched 10 hours or more?

YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

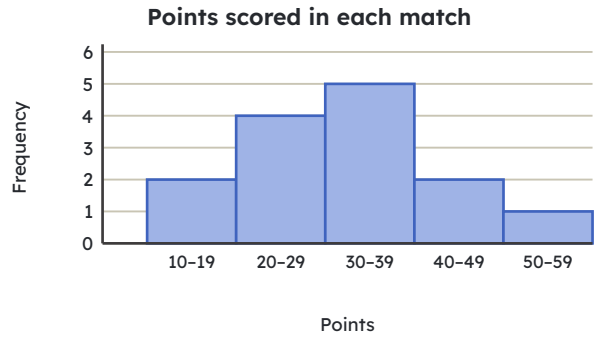
REMEMBER Read across from the top of a column to the frequency axis · Total = add up all the columns · "10 or more" = add the columns from 10 up.

ROUND 1 Questions 1–6: use the histogram of seedling heights.



1. How many seedlings are 10–19 cm?
2. How many seedlings are 30–39 cm?
3. Which class has the most seedlings?
4. Which class has the fewest seedlings?
5. How many seedlings are there in total?
6. How many seedlings are 30 cm or taller?

ROUND 2 Questions 7–12: use the histogram of points scored by a basketball team.



7. How many matches had 20–29 points?
8. How many matches had 40–49 points?
9. Which class is the most common?
10. How many matches did the team play?
11. How many matches had fewer than 30 points?
12. How many matches had 40 points or more?

Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.

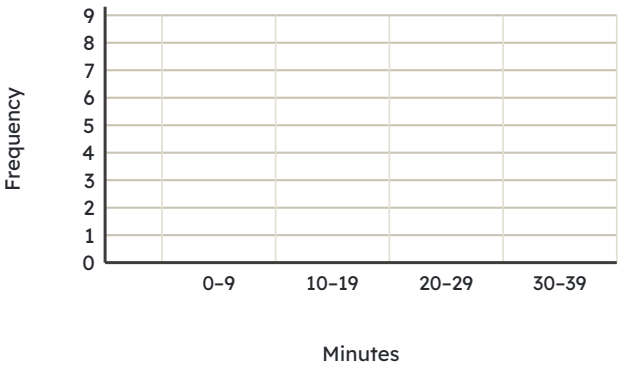
Example 2 **Minutes taken to walk to school. Percentage frequency = frequency ÷ total × 100.**

a Find the total, then the percentage frequency of each class.

Minutes	Frequency	Percentage frequency
0–9	6	
10–19	8	
20–29	4	
30–39	2	
Total		

Your teacher works this out. Copy every step.

b Draw the histogram.



YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 12 QUESTIONS

REMEMBER Percentage frequency = frequency ÷ total × 100 · Draw a histogram: label both axes, columns touch, leave a half-column gap at the start.

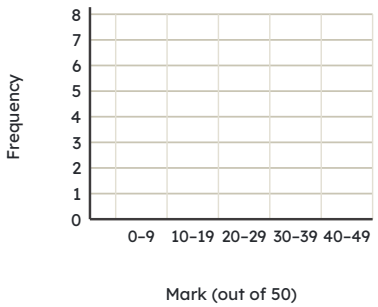
ROUND 1 Questions 1–8: find each percentage frequency (frequency ÷ total × 100).

1. 5 out of 20
2. 3 out of 10
3. 8 out of 40
4. 6 out of 25
5. 18 out of 30
6. 12 out of 50
7. 7 out of 20
8. 4 out of 16

ROUND 2 Questions 9–10: draw each histogram on its grid.

9. Draw a histogram for the table. Then name the most common class.

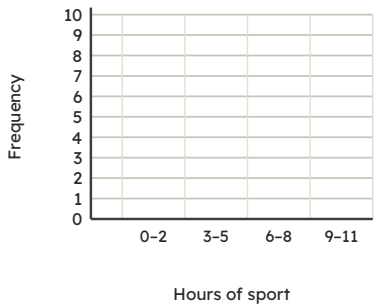
Mark	Frequency
0–9	1
10–19	3
20–29	6
30–39	7
40–49	3
Total	**20**



Most common: _____

10. Draw a histogram for the table. Then name the most common class.

Hours	Frequency
0–2	5
3–5	9
6–8	4
9–11	2
Total	**20**



Most common: _____

ROUND 3 Questions 11–12: solve each problem. Show your working.

11. 16 people were asked how many hours of TV they watch. 4 of them said 0–4 hours. What percentage is this?

12. The columns of a histogram are 3, 8, 6, 3 high. How many values are there? What percentage are in the tallest column?